Leveraging Generative AI for Interpretable Clinical Decision Making Through Causal Graphs

Mehmet Eren Ahsen[1] Rand Kittani[2] Travis Gerke[3] Laya Krishnan[2] Sean Rogan[3] Erick R. Scott[3*]

[1]Gies College of Business, UIUC [2]Carle Illinois College of Medicine, UIUC [3]cStructure, La Jolla, CA

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Abstract

Clinical AI systems' lack of interpretability limits their adoption in evidence-based medicine. To address this challenge, we propose a computational framework that harnesses generative AI's medical knowledge to create interpretable structural causal models (SCMs) for clinical decision support, quality improvement evaluation, and population health management. We evaluated our approach through a case study using data from the Midwest Healthcare Conference Causal Diagram Challenge, where we compared transformer-based large language models against human performance on a complex causal reasoning task: estimating COVID-19 treatment effects through target trial emulation. Both groups designed SCMs to evaluate glucocorticoid treatment effects on 28-day mortality using real-world data from more than 2,000 hospitalized patients, benchmarked against published RECOVERY randomized controlled trial results. The best performing SCMs achieved bootstrap coverage rates exceeding 90% for two of three severity strata. Both human and AI models demonstrated equivalent clinical plausibility (n=3 expert reviewers) and similar statistical performance, though both struggled with critical disease severity. Ablation experiments comparing SCM-based approaches against traditional potential outcomes methods revealed SCMs achieved 76-98% coverage versus 1-37% for traditional methods. These results suggest that structural causal models can effectively bridge the interpretability gap in clinical AI by providing essential scaffolding for reliable causal inference and enabling meaningful human-AI collaboration while preserving methodological rigor essential for evidence-based medicine.

Introduction: The Clinical AI Challenge

Current Limitations of Clinical AI:

- Black-box transformer models hinder clinical trust and adoption
- Post-hoc explanations (SHAP, LIME) lack causal reasoning
- FDA/regulatory agencies mandate transparency and interpretability
- Purely automated approaches cannot scale to complex clinical decisions

Our Solution:

- Convert LLM knowledge → **interpretable SCMs**
- Use **target trial emulation** to bridge real-world data and RCTs
- Enable **human-AI collaboration** via visual, debuggable causal graphs
- Provide a regulatory-compliant, transparent framework

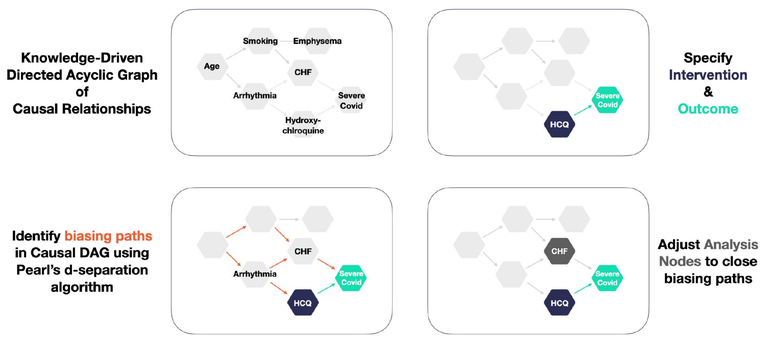


Figure 1. SCM Construction Process. Domain knowledge to causal graphs to statistical estimands.

Background: The Causal Roadmap Framework

Systematic 7-step Causal Roadmap for rigorous real-world evidence:

1. **Causal Question:** Treatment (glucocorticoids), outcome (28-day mortality), target population
2. **Observed Data:** SDY1662 quality assessment, handle missingness (LOCF), person-time format
3. **Identifiability:** Automated d-separation for confounding pathways
4. **Statistical Estimand:** Risk ratio with severity stratification
5. **Model Selection:** Pooled logistic regression, g-formula for time-varying confounding
6. **Sensitivity Analysis:** Bootstrap validation (n=500)
7. **Compare:** Human vs. AI against RECOVERY

Study Design

Participants:

- **Human Teams(n=3):** Conference participants, no formal causal training
- **LLMs (n=4):** o3, DeepSeek R1, Gemini Pro 2.5, Claude Opus

Dataset: ImmPort SDY1662

- 2,074 hospitalized COVID-19 patients (Mt. Sinai, Mar-May 2020)
- Person-time transformation: ∼65,000 person-day observations
- Severity: mild-moderate, severe, critical

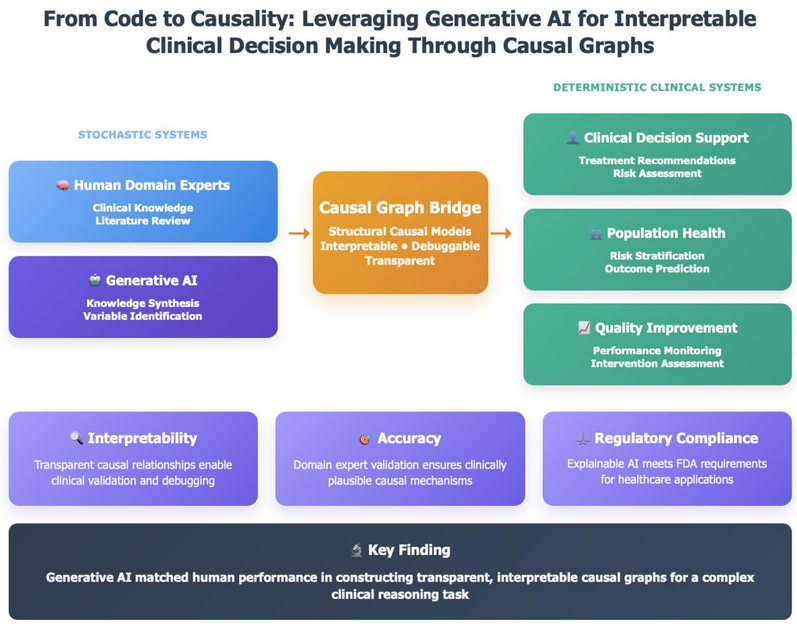
Target Trial: Glucocorticoid effects on 28-day mortality

- Benchmark: RECOVERY RCT
- Estimand: Risk ratio (RR), severity-stratified
- Method: Pooled logistic regression, g-formula
- Validation: 500 bootstrap samples/severity

Performance Metrics

Coverage: Bootstrap CI coverage vs. RECOVERY RCT (500 samples/stratum, 1,500 total)
Sign Error: Bootstrap estimates with opposite sign from RECOVERY RCT
Causal Plausibility: 3 blinded reviewers, scale 0–3 (implausible to strong evidence-based support)

Graphical Abstract



cStructure Platform

- Browser-based real-time collaboration
- Interactive DAG editing with real-time d-separation
- Privacy-preserving federated data analysis
- AI-assisted causal roadmap workflow
- Evidence integration (literature and trials)
- Graph-to-code (Python, R)

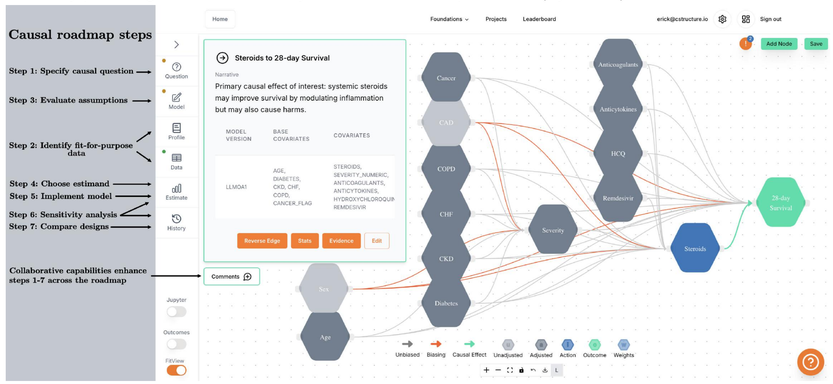


Figure 2. AI-Generated SCM. Building knowledge-based Structural Causal Models in cStructure. Creating Edges: Partici- pants could create directed edges between nodes to represent causal relationships. For example, they might add an edge from "Age" to "28-day Survival" to indicate that age directly affects survival out- comes. Adding Narratives: For each node and edge, users could provide detailed narratives explaining the node and its causal relationship with other nodes. These narratives were crucial for explaining the reasoning behind each causal connection and for evaluation of the model's plausibility. Users were encouraged to organize their graphs logically, typically with a left-to-right flow reflecting temporal precedence, where causes precede effects. This SCM along with the narrative text was generated by o3. User workflows in the cStructure platform are designed to follow the causal roadmap.

Results: Human vs. AI Performance

Model	Type	Nodes	Edges	Adj. Set Final	Plaus. Score	Outliers	Coverage (%)
o3	**LLM**	15	35	13	2.25 (0.68)	**563**	**62.4**
Team 1	**Human**	15	20	5	2.38 (0.58)	576	**61.6**
Gemini 2.5	**LLM**	13	22	4	2.59 (0.36)	580	**61.3**
Age (Base)	Base	3	3	1	3.00	604	59.7
Team 2	Human	13	16	5	2.42 (0.40)	610	59.3
Team 3	Human	6	5	4	2.27 (0.86)	612	59.2
DeepSeek R1	LLM	5	7	2	2.52 (0.42)	612	59.2
Claude Opus	LLM	19	22	9	2.29 (0.74)	640	57.3

Adj. Set Final = Adjustment Set Final, Plaus. Score = Plausibility Score (SD)

No significant difference human vs. AI ($p > 0.05$, Mann-Whitney U)
Top models: **>60% coverage** vs. RECOVERY RCT

AI: 2.41±0.16 plausibility | **Human:** 2.36±0.08
Both constructed clinically plausible causal diagrams

Bootstrap Validation & Performance Comparison

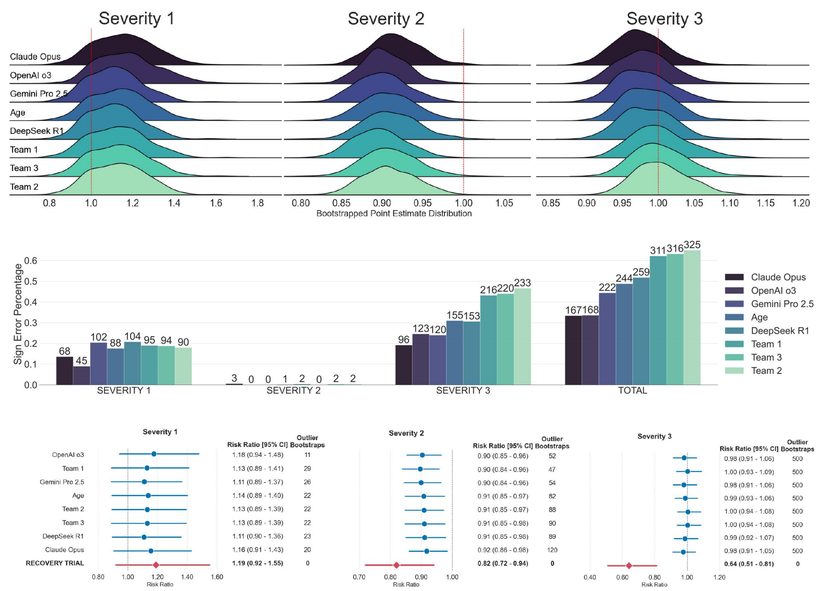


Figure 3. Top panel: Bootstrapped Risk Ratio (RR) distributions across severity levels with ridge plots showing probability densities for each SCM. Red dashed line (RR = 1.0) indicates no effect; distributions right/left suggest increased/decreased 28-day mortality risk. Models ranked by total bootstrap sign errors. **Middle panel:** Sign error distribution by model and severity. Sign errors defined as bootstrap estimates with opposite sign from corresponding RECOVERY trial point estimates (500 iterations per severity, 1,500 total per model). **Bottom panel:** Bootstrap CI coverage versus RECOVERY benchmarks. Forest plot comparing coverage performance between human participants (blue) and AI models (red) across severity strata. Points show proportion of 500 bootstrap estimates within published RECOVERY 95% CIs. Models ranked by total outliers (estimates outside RECOVERY 95% CIs); 500 samples per stratum, 1,500 total. Dashed line indicates RR = 1.

SCM Ablation: Why Causal Graphs Matter

SCMs vs. Traditional Potential Outcomes:

Model (Severity)	SCM		PO		Improvement (Δ)	
	Coverage (%)	Direction (%)	Coverage (%)	Direction (%)	Coverage (pp)	Direction (pp)
o3 (S1)	97.8	91.0	0.8	0.2	+97.0	+90.8
o3 (S2)	89.6	100.0	1.8	63.4	+87.8	+36.6
o3 (S3)	0.0	80.8	10.2	60.0	-10.2	+20.8
Claude (S1)	96.0	86.4	19.6	99.2	+76.4	-12.8
Claude (S2)	76.0	99.4	36.6	63.8	+39.4	+35.6
Claude (S3)	0.0	66.6	0.6	7.4	-0.6	+59.2
Gemini 2.5 (S1)	94.8	82.4	29.6	88.6	+65.2	-6.2
Gemini 2.5 (S2)	89.2	100.0	10.2	100.0	+79.0	0.0
Gemini 2.5 (S3)	0.0	55.6	9.8	23.2	-9.8	+32.4

RECOVERY Benchmarks: Severity 1: RR = 1.19 [0.92, 1.55] (harm) | Severity 2: RR = 0.82 [0.72, 0.94] (benefit) | Severity 3: RR = 0.64 [0.51, 0.81] (benefit).
Legend: SCM = Structural Causal Model, PO = Potential Outcomes, Δ = Improvement (pp = percentage points), S1/S2/S3 = Disease severity levels.

Why SCMs Outperform:

- **Explicit temporal structure:** Time-varying confounding encoded
- **Graphical bias detection:** D-separation identifies confounding pathways
- **Modular validation:** Each causal assumption independently testable
- **Superior estimator specification:** 76–98% coverage vs. 1–37% for PO methods

Conclusion & Future Directions

Key Findings:

- Top LLM (o3) matched best human team (62% vs. 62% coverage)
- LLMs displayed **superior directional accuracy** — critical for early decisions
- **SCMs dramatically outperform PO framework** (76–98% vs. 1–37%)
- Rregulatory-aligned pathway for clinical AI
- Visual causal graphs enabled effective human-AI collaboration

Future Directions:

- Expand to **oncology, cardiology, and immunology**
- Evaluate accuracy of **individual-level treatment predictions**
- Integrate Shapley Flow for causal-predictive model interpretation
- Health system **quality improvement integration**
- Partner with regulatory agencies for evidence generation standards

SCMs Bridge AI Knowledge and Human Oversight for Trustworthy Clinical Reasoning

References & Links

Key References:

Hernán MA, Robins JM. Using Big Data to Emulate a Target Trial. *Am J Epidemiol.* 2016.
Pearl J. *Causality: Models, Reasoning, and Inference.* 2nd ed. Cambridge Univ. Press, 2009.
RECOVERY Trial. Dexamethasone in COVID-19. *N Engl J Med.* 2021.

Data: ImmPort.org (SDY1662) | **Code:** github.com/cstructure/interpretable-clinical-scm

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